Prof. Nicanor Quijano Ph.D., IEEE Senior Member

Full Professor, Electrical and Electronics Eng.
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Profile

Control and automation researcher with close to twenty years of faculty experience, working on hierarchical and distributed optimization and control for large-scale multi-agent systems, using learning, bio-inspired, and game-theoretical techniques. The methods are developed with, and validated on, automation problems in energy systems and microgrids, water distribution and urban drainage, open-channel irrigation and smart agriculture, bioprocess and manufacturing scheduling, multi-robot and UAV coordination, and transportation. Author of more than 130 peer-reviewed publications, three books, and eight book chapters; more than 5,000 citations. Eleven Ph.D. theses completed, two of them recognized with European best-thesis awards (2018, 2025). Nine years of continuous service as Associate Editor of three international journals, including the IEEE Transactions on Control Systems Technology.

Education

Ph.D. in Electrical and Computer Engineering 2006
The Ohio State University, Columbus, Ohio, USA — Major: Controls and Signal Processing
Dissertation: *The Ideal Free Distribution: Theory and Engineering Application*. Advisor: Prof. Kevin M. Passino.

M.S. in Electrical Engineering 2002
The Ohio State University, Columbus, Ohio, USA — Major: Controls and Signal Processing
Thesis: *Experiments and Technologies for Decentralized Temperature Control*. Advisor: Prof. Kevin M. Passino.

B.S. in Electronics Engineering 1999
Pontificia Universidad Javeriana, Bogotá, Colombia
Project: *Simulador digital para control y evaluación de procesos industriales*. Advisor: Prof. Asdrúbal Espitia.

Academic Appointments

Full Professor 2015 – present
Dept. of Electrical and Electronics Engineering, Universidad de los Andes, Bogotá, Colombia

Director, Research Group in Control and Automation Systems (GIAP) 2007 – present
Universidad de los Andes, Bogotá, Colombia

Associate Professor 2010 – 2015
Dept. of Electrical and Electronics Engineering, Universidad de los Andes

Assistant Professor 2007 – 2010
Dept. of Electrical and Electronics Engineering, Universidad de los Andes

Graduate Research and Teaching Associate 2001 – 2006
Dept. of Electrical and Computer Engineering, The Ohio State University

Instructor and Teaching Assistant 1996 – 2000
Pontificia Universidad Javeriana, Bogotá, Colombia

Editorial Experience and Peer Review

Associate editorships

- **IEEE Transactions on Control Systems Technology** — Associate Editor, 2017–2023.
- **Energy Systems** (Springer) — Associate Editor, 2018–2023.
- **Journal of Modern Power Systems and Clean Energy** — Associate Editor, 2016–2022.

- **Artificial Intelligence Research** (AIR) — Editorial Board member.

Guest editorial work

- Guest Editor, Special Issue on *Game-Theoretical Methods in Control of Engineering Systems*, IEEE Control Systems Magazine, vol. 37, no. 1, 2017 (with C. Ocampo-Martinez); co-author of the editorial introduction to the special issue.
- Co-editor, *Energy Systems Integration for Multi-Energy Systems: From Operation to Planning in the Green Energy Context*, Springer Nature, 2025 [B2].

Reviewing

- Regular reviewer for, among others: IEEE Transactions on Automatic Control; Automatica; IEEE Transactions on Control Systems Technology; IEEE Transactions on Control of Network Systems; IEEE Transactions on Smart Grid; IEEE Transactions on Systems, Man, and Cybernetics; IEEE Transactions on Evolutionary Computation; IEEE Control Systems Letters; International Journal of Electrical Power and Energy Systems; Revista Ingeniería (Uniandes).
- Conference reviewer for IEEE CDC, IEEE ACC, ECC, IFAC World Congress, IEEE CCTA, IEEE CEC, and IEEE CCAC.
- Ph.D. thesis examiner and jury member for universities in Colombia, Spain, Belgium, and France.

PROFESSIONAL SERVICE AND LEADERSHIP

- Member, Board of Governors, IEEE Control Systems Society (CSS), 2014.
- Chair, IEEE Control Systems Society, Colombia Chapter, 2011–2013.
- Member, IEEE Technical Committee on Smart Grids (CSS and IES); Lead of the sub-technical committee on microgrids, IEEE CSS, 2011–2017.
- Publicity Chair, IEEE Conference on Control Technology and Applications (CCTA), 2027.
- International Program Committee: IEEE CDC 2027, IEEE CDC 2025, IFAC SENSYS 2025, IFAC COSY 2025, IFAC CLCA 2025, IEEE ACC 2019, IFAC NOLCOS 2016, IEEE CDC 2011.
- Organizing committee, IEEE Colombian Conference on Automatic Control (CCAC): 2011, 2015, 2017, 2019, 2021, 2023.
- Organizer of the poster session, International Microgrid Symposium 2026.
- Member of the Colciencias committee that defined national research directions in electronics for the Colombian Ministry of Defense, 2009.
- Senior Member of the IEEE since 2013 (IEEE Number: 41529030).

AWARDS AND HONORS

- **European Systems & Control Ph.D. Thesis Award, 2025** — awarded to Juan Martínez-Piazuelo, co-advised with Prof. C. Ocampo-Martínez, for the thesis *On the Design of Optimization-Based Controllers via Generalized Nash Equilibrium Seeking in Evolutionary Games over Networks*.
- **Alain Gauthier Award, 2025** — IEEE Control Systems Society, Colombia.
- **Experienced Research Award, 2021** — School of Engineering, Universidad de los Andes.
- **ISA Transactions Best Paper Award, 2018** — for *Dynamical Tuning for MPC Using Population Games: A Water Supply Network Application*.
- **Ph.D. Award on Control for Complex and Heterogeneous Systems, Europe 2018** — awarded to Julián Barreiro-Gómez, co-advised with Prof. C. Ocampo-Martínez.
- IEEE Senior Member, 2013.
- Distinguished Lecturer, School of Engineering, Universidad de los Andes, 2008.

RESEARCH FUNDING

Selected participation as principal investigator (PI) or co-PI:

- **PI (Uniandes)** — *Bridging Europe and Latin America: Fostering Student Mobility and Collaborative Learning between Universidad de los Andes, CentraleSupélec, and UMONS*, T.I.M.E. Project 2026.
- **PI (Uniandes)** — *Desarrollo de un modelo transaccional de energía no convencional de múltiples agentes en el departamento de Nariño*, Sistema General de Regalías (SGR) Nariño, 2022–2026.

- **PI** — *Systemic Perspectives on Low-Carbon Cities in Colombia: An Integrated Urban Modelling Approach for Policy and Regulatory Analysis*, UK-PACT Colombia, 2019–2021.
- **Co-PI (Uniandes)** — *DDDAS Anomaly Detection and Response for Attack-Resilient Multi-Agent Systems*, SOARD/AFOSR, USA, 2018–2021.
- **PI** — *Feasibility Study for Energy Storage*, EPRI–Uniandes, 2018–2019.
- **PI** — *Solución Energética Piloto La Guajira*, ISAGEN–Uniandes, 2016–2019.
- **Co-PI** — *Uso de energía eléctrica en población vulnerable*, SGR Cundinamarca (Uniandes, UNal, PUJ, CIF), 2015–2019.
- **Co-PI** — *Análisis de oportunidades energéticas con fuentes alternativas*, SGR Nariño (UdeNar, Uniandes), 2014–2016.
- **PI** — *Drenaje urbano y cambio climático: hacia los sistemas de drenaje urbano*, Phases I–II, Colciencias–Pavco/Mexichem, 2012–2015.
- **Co-PI** — *SILICE*, Phases 1–3, Colciencias–Codensa (Uniandes, UNal, PUJ, CIDET, UIS), 2008–2015.

PUBLICATIONS

More than 130 peer-reviewed publications; more than 5,000 citations (Google Scholar). Complete and updated list: Google Scholar profile.

Books

[B1] J. Martinez-Piazuelo, C. Ocampo-Martinez, and **N. Quijano**, *Generalized Nash Equilibrium Seeking in Population Games: Applications in the Control of Large-Scale Multi-agent Systems*, Communications and Control Engineering. Springer Nature, 2026.

[B2] C. Ocampo-Martinez and **N. Quijano** (Eds.), *Energy Systems Integration for Multi-Energy Systems: From Operation to Planning in the Green Energy Context*. Springer Nature, 2025.

[B3] R. Smith-Quintero, A.Cadena, M.Espinosa, and **N. Quijano**, *Ciudades sostenibles: Un enfoque de modelaje urbano integrado para el análisis de política en Colombia*. Ediciones Uniandes, 2022.

[B4] **N. Quijano**, J. Barreiro-Gómez, G. Obando, J. Martínez-Piazuelo, and C. Ocampo-Martínez, *Introducción a la teoría de juegos: de lo clásico a lo evolutivo, llevándolo a la ingeniería*. In preparation.

Book chapters

[BC1] A. Cusva, G. Jiménez-Estévez, R. Sierra-Ramírez, and **N. Quijano**, "Integrated multi-energy systems analysis for a Colombian cocoa plantation," in *Energy Systems Integration for Multi-Energy Systems*, Springer Nature, 2025.

[BC2] D. Erazo-Caicedo, **N. Quijano**, C. Ocampo-Martinez, and G. Jiménez-Estévez, "Integrated energy systems: an overview from a multi-layer architecture," in *Energy Systems Integration for Multi-Energy Systems*, Springer Nature, 2025.

[BC3] L. F. Cómbita, J. Giraldo, A. Cárdenas, and **N. Quijano**, "DDDAS for attack detection and isolation of control systems," in *Handbook of Dynamic Data Driven Applications Systems*, pp. 407–422, Springer-Verlag, 2018.

[BC4] J. Barreiro-Gómez, G. Riaño-Briceño, C. Ocampo-Martínez, and **N. Quijano**, "Data-driven evolutionary-game based control for drinking-water networks," in *Real-Time Monitoring and Operational Control of Drinking-Water Systems*, pp. 363–383, Springer-Verlag, 2017.

[BC5] J. Barreiro-Gómez, C. Ocampo-Martínez, and **N. Quijano**, "Evolutionary-game-based dynamical tuning for multi-objective model predictive control," in *Developments in Model-Based Optimization and Control*, pp. 115–138, Springer-Verlag, 2015.

[BC6] J. Giraldo, **N. Quijano**, and K. M. Passino, "Honey bee social foraging algorithm for resource allocation," in *Springer Handbook of Computational Intelligence*, pp. 1361–1376, Springer-Verlag, 2015.

[BC7] A. Pantoja, **N. Quijano**, and G. Obando, "Bioinspired approach for a multizone temperature control," in *Handbook of Biomimetics and Bioinspiration*, pp. 817–860, World Scientific, 2014.

[BC8] C. Barreto, A. Cárdenas, and **N. Quijano**, "Controllability of dynamical systems: threat models and reactive security," in *Decision and Game Theory for Security*, pp. 45–64, Springer-Verlag, 2013.

[BC9] M. Amin, A. Annaswamy, A. Cadena, D. Callaway, M. Caramanis, J. Chow, D. Dotta, A. Farid, P. Flikkema, S. Genc, S. Low, E. Mojica-Nava, A. Parisio, M. Polis, Z. Qu, **N. Quijano**, and T. Samad, "Control-enabled smart grids scenarios for 2030–2050," in *IEEE Vision for Smart Grid Controls: 2030 and Beyond*, IEEE, 2013.

[BC10] **N. Quijano**, J. Finke, and K. M. Passino, "Laboratorios de bajo costo," in *Ciencia e ingeniería en la formación de ingenieros para el siglo XXI*, pp. 171–188, ACOFI, 2008.

Journal articles

[J1] V.Vargas-Panesso, **N. Quijano**, and L.F.Giraldo, "Emergence of contagious cooperation in community-structured networks," *Phys. Rev. E*, vol 114(3), art 034309, 2026.

[J2] D.Erazo-Caicedo, S.Olaru, P.Panciatici, J.Revelo-Fuelagán, **N. Quijano**, and G.Jiménez-Estévez, "A novel market-clearing dynamic system for scalable bipartite local energy markets," *Control Engineering Practice*, vol 173, art 106962, 2026.

[J3] M.Ríos, R.Manrique, **N. Quijano**, and L.F.Giraldo, "The Illusion of Rationality: Tacit Bias and Strategic Dominance in Frontier LLM Negotiation Games," *IEEE Transactions on Artificial Intelligence*, pp.1-8, 2026.

[J4] J. Martinez-Piazuelo, C. Ocampo-Martinez, and **N. Quijano**, "A delta-dissipativity-based framework for generalized Nash equilibrium seeking in population games," *Automatica*, vol. 187, art. 112909, 2026.

[J5] CK.Dinh, V.Marguet, I.Prodan, C.Ocampo-Martinez, **N. Quijano**, and J.Barreiro-Gomez, "B-splines-based mechanism design in population games for distributed evolutionary dynamics formation control," *Control Engineering Practice*, vol.168, art. 106676, 2026.

[J6] J.Caballé, P.Segovia, C.Ocampo-Martinez, and **N. Quijano**, "Large-scale EV charging coordination: a detailed exploration of mean-field and reinforcement learning approaches," *Control Engineering Practice*, vol.168, art. 106669, 2026.

[J7] A.Cusva-García, G.Jiménez-Estévez, and **N. Quijano**, "A methodological framework for the optimization of agro-energy systems within the water–energy–food nexus," *Energy Nexus*, art. 100653, 2026.

[J8] V.Vargas-Panesso, **N. Quijano**, L.F.Giraldo, and J.Barreiro-Gomez, "Modeling Strategic Intercommunity Connections in Evolutionary Games," *IEEE Transactions on Cybernetics*, vol.56(6), pp.3074-3087, 2026.

[J9] M. Mosquera, J. S. Pinzón, Y. Fonseca, M. Ríos, **N. Quijano**, L. F. Giraldo, and R. Manrique, "Can LLM-augmented autonomous agents cooperate? An evaluation of their cooperative capabilities through Melting Pot," *IEEE Transactions on Artificial Intelligence*, vol. 7, no. 2, pp. 674–683, 2026.

[J10] M. Chacón-Chamorro, L. F. Giraldo, **N. Quijano**, V. Vargas-Panesso, C. González, J. S. Pinzón, R. Manrique, M. Ríos, Y. Fonseca, D. F. Gómez-Barrera, and M. T. Perdomo, "Cooperative resilience in artificial intelligence multiagent systems," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 12, pp. 3430–3440, 2025.

[J11] G. J. Conde, C. Ocampo-Martinez, and **N. Quijano**, "A novel nonlinear model predictive control approach to minimize losses in open-channel irrigation systems," *ISA Transactions*, vol.159, pp.278-292, 2025.

[J12] J. Martinez-Piazuelo, C. Ocampo-Martinez, and **N. Quijano**, "Distributed Nash equilibrium seeking in strongly contractive aggregative population games," *IEEE Transactions on Automatic Control*, vol. 69, no. 7, pp. 4427–4442, 2024.

[J13] J. Lopez-Jimenez, **N. Quijano**, and A. Vande Wouwer, "Agent-based sensor location strategy for smart irrigation of large crop fields," *Computers and Electronics in Agriculture*, vol. 214, art. 108282, 2023.

[J14] J. Martinez-Piazuelo, W. Ananduta, C. Ocampo-Martinez, S. Grammatico, and **N. Quijano**, "Population games with replicator dynamics under event-triggered payoff provider and a demand response application," *IEEE Control Systems Letters*, vol. 7, pp. 3417–3422, 2023.

[J15] A. Sánchez-Amores, J. Martinez-Piazuelo, J. M. Maestre, C. Ocampo-Martinez, E. F. Camacho, and **N. Quijano**, "Coalitional model predictive control of parabolic-trough solar collector fields with population-dynamics assistance," *Applied Energy*, vol. 334, art. 120740, 2023.

[J16] L. F. Cómbita, **N. Quijano**, and A. Cárdenas, "Defending state-feedback based controllers against sensor attacks," *Ingeniería*, vol. 28, no. 2, e20094, 2023.

[J17] D. Gómez, **N. Quijano**, and L. F. Giraldo, "Information optimization and transferable state abstractions in deep reinforcement learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.

[J18] J. Martinez-Piazuelo, **N. Quijano**, and C. Ocampo-Martinez, "Nash equilibrium seeking in full-potential population games under capacity and migration constraints," *Automatica*, vol. 141, art. 110285, 2022.

[J19] J. Martinez-Piazuelo, **N. Quijano**, and C. Ocampo-Martinez, "A payoff dynamics model for generalized Nash equilibrium seeking in population games," *Automatica*, vol. 140, art. 110227, 2022.

[J20] J. Martinez-Piazuelo, C. Ocampo-Martinez, and **N. Quijano**, "On distributed Nash equilibrium seeking in a class of contractive population games," *IEEE Control Systems Letters*, vol. 6, pp. 2972–2977, 2022.

[J21] G. Obando, **N. Quijano**, and C. Ocampo-Martinez, "Decentralized control for urban drainage systems using replicator dynamics," *IEEE Access*, vol. 10, pp. 56740–56762, 2022.

[J22] L. F. Cómbita, **N. Quijano**, and Á. A. Cárdenas, "On the stability of cyber-physical control systems with sensor multiplicative attacks," *IEEE Access*, vol. 10, pp. 39716–39728, 2022.

[J23] J. Lopez-Jimenez, A. Vande Wouwer, and **N. Quijano**, "Dynamic modeling of crop–soil systems to design monitoring and automatic irrigation processes: a review with worked examples," *Water*, vol. 14, no. 6, art. 889, 2022.

[J24] J. Martinez-Piazuelo, G. Diaz-Garcia, **N. Quijano**, and L. F. Giraldo, "Discrete-time distributed population dynamics for optimization and control," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 2022.

[J25] J. Martinez-Piazuelo, **N. Quijano**, and C. Ocampo-Martinez, "Decentralized charging coordination of electric vehicles under feeder capacity constraints," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 4, pp. 1600–1610, 2021.

[J26] G. Conde, **N. Quijano**, and C. Ocampo-Martinez, "Modeling and control in open-channel irrigation systems: a review," *Annual Reviews in Control*, vol. 51, pp. 153–171, 2021.

[J27] G. Conde, **N. Quijano**, and C. Ocampo-Martinez, "Detection, isolation, and magnitude estimation of unknown flows in open-channel irrigation systems," *IEEE Access*, vol. 9, pp. 115348–115369, 2021.

[J28] J. Barreiro-Gomez, I. Mas, J. I. Giribet, P. Moreno, C. Ocampo-Martinez, R. Sánchez-Peña, and **N. Quijano**, "Distributed data-driven UAV formation control via evolutionary games: experimental results," *Journal of the Franklin Institute*, vol. 358, no. 10, pp. 5334–5352, 2021.

[J29] M. A. Velasquez, **N. Quijano**, A. I. Cadena, and M. Shahidehpour, "Distributed stochastic economic dispatch via model predictive control and data-driven scenario generation," *International Journal of Electrical Power and Energy Systems*, vol. 129, art. 106796, 2021.

[J30] J. Martinez-Piazuelo, **N. Quijano**, and C. Ocampo-Martinez, "A payoff dynamics model for equality-constrained population games," *IEEE Control Systems Letters*, vol. 6, pp. 530–535, 2021.

[J31] L. Burbano, L. F. Cómbita, **N. Quijano**, and S. Rueda, "Dynamic data integration for resilience to sensor attacks in multi-agent systems," *IEEE Access*, vol. 9, pp. 31236–31245, 2021.

[J32] J. Lopez-Jimenez, **N. Quijano**, and A. Vande Wouwer, "An agent-based crop model framework for heterogeneous soils," *Agronomy*, vol. 11, no. 1, art. 85, 2021.

[J33] N. España, J. Barco-Jiménez, A. Pantoja, and **N. Quijano**, "Distributed population dynamics for active and reactive power dispatch in islanded microgrids," *International Journal of Electrical Power and Energy Systems*, vol. 125, art. 106407, 2021.

[J34] M. Velasquez, J. Barreiro-Gomez, **N. Quijano**, A. Cadena, and M. Shahidehpour, "Intra-hour microgrid economic dispatch based on model predictive control," *IEEE Transactions on Smart Grid*, vol. 11, no. 3, pp. 1968–1979, 2020.

[J35] J. Martinez-Piazuelo, D. E. Ochoa, **N. Quijano**, and L. F. Giraldo, "A multi-critic reinforcement learning method: an application to multi-tank water systems," *IEEE Access*, vol. 8, pp. 173227–173238, 2020.

[J36] D. E. Ochoa, J. I. Poveda, C. A. Uribe, and **N. Quijano**, "Robust optimization over networks using distributed restarting of accelerated dynamics," *IEEE Control Systems Letters*, vol. 5, no. 1, pp. 301–306, 2020.

[J37] L. Cómbita, A. Cárdenas, and **N. Quijano**, "Mitigation response to sensor attacks on cyber-physical control systems," *IEEE Access*, vol. 7, pp. 92444–92455, 2019.

[J38] M. Velasquez, J. Barreiro-Gomez, **N. Quijano**, A. Cadena, and M. Shahidehpour, "Distributed model predictive control for economic dispatch of power systems with high penetration of renewable energy resources," *International Journal of Electrical Power and Energy Systems*, vol. 113, no. 1, pp. 607–617, 2019.

[J39] J. Giraldo, E. Mojica-Nava, and **N. Quijano**, "Synchronisation of heterogeneous Kuramoto oscillators with sampled information and a constant leader," *International Journal of Control*, vol. 92, no. 11, pp. 2591–2607, 2019.

[J40] M. Velásquez, O. Torres-Pérez, **N. Quijano**, and A. Cadena, "Hierarchical dispatch of multiple microgrids by using nodal prices: approaches from consensus and replicator dynamics," *Journal of Modern Power Systems and Clean Energy*, vol. 7, no. 4, pp. 1–12, 2019.

[J41] J. Barreiro-Gómez, G. Obando, C. Ocampo-Martínez, and **N. Quijano**, "An evolutionary-games approach for distributed predictive control involving resource allocation," *IET Control Theory and Applications*, vol. 13, no. 6, pp. 772–782, 2019.

[J42] J. Barreiro-Gomez, F. Bianchi, C. Ocampo-Martínez, and **N. Quijano**, "Data-driven decentralized algorithm for wind farms control with population-games assistance," *Energies*, vol. 12, no. 6, art. 1164, 2019.

[J43] C. Barreto, E. Mojica-Nava, and **N. Quijano**, "Incentive mechanisms to prevent efficiency loss in regulated markets," *International Journal of Electrical Power and Energy Systems*, vol. 110, pp. 523–535, 2019.

[J44] J. Barreiro-Gomez, C. Ocampo-Martinez, and **N. Quijano**, "Time-varying partitioning for predictive control design: density-games approach," *Journal of Process Control*, vol. 75, pp. 1–14, 2019.

[J45] A. Pantoja, **N. Quijano**, and G. Obando, "Distributed optimization with information-constrained population dynamics," *Journal of the Franklin Institute*, vol. 356, no. 1, pp. 209–236, 2019.

[J46] **N. Quijano**, A. Pedraza, M. Velásquez, G. Jiménez-Estévez, Á. Cadena, J. M. Becerra, and Á. Ramírez, "Microrredes aisladas en La Guajira: diseño e implementación," *Revista de Ingeniería, Universidad de los*

Andes, vol. 48, pp. 54–65, 2019.

[J47] W. Ananduta, J. Barreiro-Gomez, C. Ocampo-Martinez, and **N. Quijano**, "On the mitigation of communication failures in distributed model predictive control strategies," *IET Control Theory and Applications*, vol. 12, no. 18, pp. 2507–2515, 2018.

[J48] A. Rengifo, F. E. Segura-Quijano, and **N. Quijano**, "An affordable set of control system laboratories using a low-cost robotic platform," *IEEE/ASME Transactions on Mechatronics*, vol. 23, no. 4, pp. 1705–1715, 2018.

[J49] **N. Quijano**, C. Ocampo-Martinez, J. Barreiro-Gomez, G. Obando, A. Pantoja, and E. Mojica-Nava, "The role of population games and evolutionary dynamics in distributed control systems," *IEEE Control Systems Magazine*, vol. 37, no. 1, pp. 70–97, 2017.

[J50] C. Ocampo-Martinez and **N. Quijano**, "Game-theoretical methods in control of engineering systems: an introduction to the special issue," *IEEE Control Systems Magazine*, vol. 37, no. 1, pp. 30–32, 2017.

[J51] J. Barreiro-Gómez, C. Ocampo-Martínez, and **N. Quijano**, "Dynamical tuning for MPC using population games: a water supply network application," *ISA Transactions*, vol. 69, pp. 175–186, 2017. *(ISA Transactions Best Paper Award)*

[J52] J. Giraldo, A. Cárdenas, and **N. Quijano**, "Integrity attacks on real-time pricing in smart grids: impact and countermeasures," *IEEE Transactions on Smart Grid*, vol. 8, no. 5, pp. 2249–2257, 2017.

[J53] J. Barreiro-Gómez, G. Obando, and **N. Quijano**, "Distributed population dynamics: optimization and control applications," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 47, no. 2, pp. 304–314, 2017.

[J54] J. Barreiro-Gómez, C. Ocampo-Martínez, **N. Quijano**, and J. M. Maestre, "Non-centralized control for flow-based distribution networks: a game-theoretical insight," *Journal of the Franklin Institute*, vol. 354, no. 9, pp. 1830–1845, 2017.

[J55] G. Obando, **N. Quijano**, and N. Rakoto-Ravalontsalama, "A center-free approach for resource allocation with lower bounds," *International Journal of Control*, vol. 90, no. 9, pp. 1830–1845, 2017.

[J56] E. Mojica-Nava, S. Rivera, and **N. Quijano**, "Game-theoretic dispatch control in microgrids considering network losses and renewable distributed energy resources integration," *IET Generation, Transmission and Distribution*, vol. 11, no. 6, pp. 1583–1590, 2017.

[J57] P. Ñañez, R. G. Sanfelice, and **N. Quijano**, "On an invariance principle for differential-algebraic equations with jumps and its application to switched differential-algebraic equations," *Mathematics of Control, Signals, and Systems*, vol. 29, no. 1, art. 5, 2017.

[J58] M. Velásquez, **N. Quijano**, and A. Cadena, "Model predictive control applied to the dynamic economic dispatch problem," *IEEE Latin America Transactions*, vol. 15, no. 4, pp. 656–662, 2017.

[J59] J. Barco, A. Guerra, L. Muñoz, and **N. Quijano**, "Optimal routing and scheduling of charge for electric vehicles: a case study," *Mathematical Problems in Engineering*, 2017.

[J60] J. Barreiro-Gómez, **N. Quijano**, and C. Ocampo-Martínez, "Constrained distributed optimization: a population dynamics approach," *Automatica*, vol. 69, pp. 101–116, 2016.

[J61] G. Riaño-Briceño, J. Barreiro-Gómez, A. Ramírez-Jaime, **N. Quijano**, and C. Ocampo-Martínez, "MatSWMM: an open-source toolbox for designing real-time control of urban drainage systems," *Environmental Modelling and Software*, vol. 83, pp. 143–154, 2016.

[J62] G. Obando, J. Poveda, and **N. Quijano**, "Replicator dynamics under perturbations and time delays," *Mathematics of Control, Signals, and Systems*, vol. 28, no. 3, art. 20, 2016.

[J63] M. Velásquez, **N. Quijano**, and A. Cadena, "Optimal placement of switches on DG enhanced feeders with short circuit constraints," *Electric Power Systems Research*, vol. 141, pp. 221–232, 2016.

[J64] A. M. Ospina and **N. Quijano**, "Distributed control of small-scale power systems using noncooperative games," *International Journal of Electrical Power and Energy Systems*, vol. 82, no. 1, pp. 535–544, 2016.

[J65] L. García, J. Barreiro-Gómez, E. Escobar, D. Téllez, **N. Quijano**, and C. Ocampo-Martínez, "Modeling and real-time control of urban drainage systems: a review," *Advances in Water Resources*, vol. 85, pp. 120–132, 2015.

[J66] E. Mojica-Nava, C. Barreto, and **N. Quijano**, "Population games methods for distributed control of microgrids," *IEEE Transactions on Smart Grid*, vol. 6, no. 6, pp. 2586–2595, 2015.

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polynomial systems using dissipation inequalities," in *Proc. IEEE Conference on Decision and Control (CDC)*, pp. 31–36, 2008.

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[C108] J. Finke, **N. Quijano**, and K. M. Passino, "Ideal free distribution growing on networks," in *Proc. American Control Conference (ACC)*, pp. 159–164, 2008.

[C109] C. Ramirez, I. Gil, and **N. Quijano**, "Modeling and control of a packed distillation column," in *Proc. AIChE Annual Meeting*, 2008.

[C110] **N. Quijano** and K. M. Passino, "Honey bee social foraging algorithms for resource allocation, Part I: algorithm and theory," in *Proc. American Control Conference (ACC)*, pp. 3383–3388, 2007.

[C111] **N. Quijano** and K. M. Passino, "Honey bee social foraging algorithms for resource allocation, Part II: application," in *Proc. American Control Conference (ACC)*, pp. 3389–3394, 2007.

[C112] **N. Quijano** and K. M. Passino, "Optimality and stability of the ideal free distribution with application to temperature control," in *Proc. American Control Conference (ACC)*, pp. 4837–4842, 2006.

[C113] **N. Quijano** and K. M. Passino, "Resource allocation strategies for multizone temperature control," in *Proc. IFAC Symposium on System, Structure and Control*, 2004.

National conference proceedings (selected)

Sixteen additional papers published in Colombian national conferences (Congreso de la Asociación Colombiana de Automática, Colombian IEEE Workshop on Robotics and Automation, 2007–2009), on bio-inspired and evolutionary-game methods for temperature control, urban traffic, water distribution, hybrid systems, and cancer therapy. Full list available on request.

ADVISING AND MENTORING

- **Doctoral:** 11 Ph.D. theses completed as advisor or co-advisor; currently co-advising three Ph.D. students. Two theses received European best-thesis awards (Julián Barreiro-Gómez, 2018; Juan Martínez-Piazuelo, 2025).
- **Postdoctoral:** three postdoctoral researchers supervised.
- **Master's:** 35 M.Sc. theses completed; currently advising three.
- **Undergraduate:** more than 40 bachelor theses and capstone projects.
- Students placed in academic positions and research centers in Colombia, Spain, Belgium, the United States, and the United Arab Emirates.

TEACHING

- **Graduate:** Nonlinear Systems and Control; Intelligent and Process Control; Population Dynamics and Evolutionary Games; Learning and Evolution Oriented to Control; Microgrids and Energy Communities; Model Predictive Control (2007–present).
- **Undergraduate:** Automatic Control; Modeling and Analysis of Control Systems; Instrumentation and Control Technology; Dynamical Systems; Capstone Project (2007–present).
- **Massive open online course:** lead instructor of *Introducción al análisis de sistemas de control con MATLAB* (Coursera, in Spanish), developed with MathWorks.
- Development of low-cost and remote control laboratories, and of flipped-classroom materials for control education.

INVITED TALKS, PLENARIES, AND SHORT COURSES (SELECTED)

- Invited workshop speaker, IFAC World Congress, 2026.
- Invited plenary, 6th IEEE Colombian Conference on Automatic Control (CCAC), 2023.
- Invited talk, MathWorks, *Introducción al análisis de sistemas de control con MATLAB*, 2023.
- Invited plenary, XVII Expoconstrucción–Expodiseño, 2023.
- Invited panelist, 1er Congreso Energía y Transición Energética, 2023.
- Invited plenary, 41st Benelux Meeting on Systems and Control, Brussels, 2022.
- Invited talk, University of California, Santa Cruz, 2022.

- Plenary address, Senate of Colombia, *Transformación energética*, 2021, and *Seguridad energética*, 2020.
- Invited talk, *Predictive Control Meets Evolutionary Games*, Universidad de Almería, 2020.
- Invited panelist on control education, IFAC World Congress, 2020, and American Control Conference, 2019.
- Short course on game theory, INCITE, Barcelona, 2017.
- Short course on Model Predictive Control and Population Games, IFAC Latin-American Control Conference, Medellín, 2016.
- More than 25 additional plenary and invited talks at universities and technical events in Colombia, Chile, Spain, and the United States (2008–2019).

SKILLS AND LANGUAGES

- **Engineering tools:** MATLAB/Simulink, LabVIEW, dSPACE, CasADi, SWMM/MatSWMM, LaTeX.
- **Programming:** Python, C, MATLAB scripting.
- **Hardware:** dSPACE 1104 controller board, National Instruments DAQ, PIC and 8051-family microcontrollers.
- **Languages:** Spanish (native), English (fluent, written and oral), French (fluent, written and oral).

Full publication record.
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